

Lecture 1: The Engineering Context of Nuclear Power: Notes

Intro to Nuclear Engineering – D. T. Leighton

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1 Electricity Requirements and Projections

- **Current US Generation (2024):** Total net generation reached approximately **4,401 TWh**. This marks a significant 3% year-over-year increase, the fifth-highest growth rate this century ([Ember/EIA 2025 Report](#)).
- **The Historical Plateau (2008–2022):** For nearly 15 years, US demand remained stagnant with an average growth rate of only **0.1% per year**, as efficiency gains (e.g., LED lighting) offset economic growth ([EIA, May 2025](#)).
- **The 2024–2030 Surge:** Demand is now projected to grow by **1.7% to 2.4% annually** through 2030, a sharp departure from the previous decade ([EIA STEO, Dec 2025](#)).
- **Growth Driver: Data Centers & AI**
 - In 2023, data centers consumed **176 TWh** (~4.4% of US load).
 - Consumption is projected to reach **6.7% to 12%** of total US electricity by 2028 ([LBNL 2024 Report](#)).
 - EPRI estimates that AI-specific queries require **10–20x** the power of a standard search ([DOE 2024/2025 Resources](#)).
- **Growth Driver: Transportation & Industry**
 - Light-duty EV demand reached **11 TWh** in 2024 and is scaling at a rate of 8%+ per year.
 - The "Reshoring" of semiconductor and battery fabrication is creating localized "Gigawatt-scale" demand clusters ([Grid Strategies 2024](#)).
- **Long-term Projection (2050):** To meet economy-wide Net-Zero goals, the IEA and DOE forecast a **doubling** of current electricity demand to ~8,000 TWh to replace direct combustion in heating and transit ([IEA WEO 2024](#)).

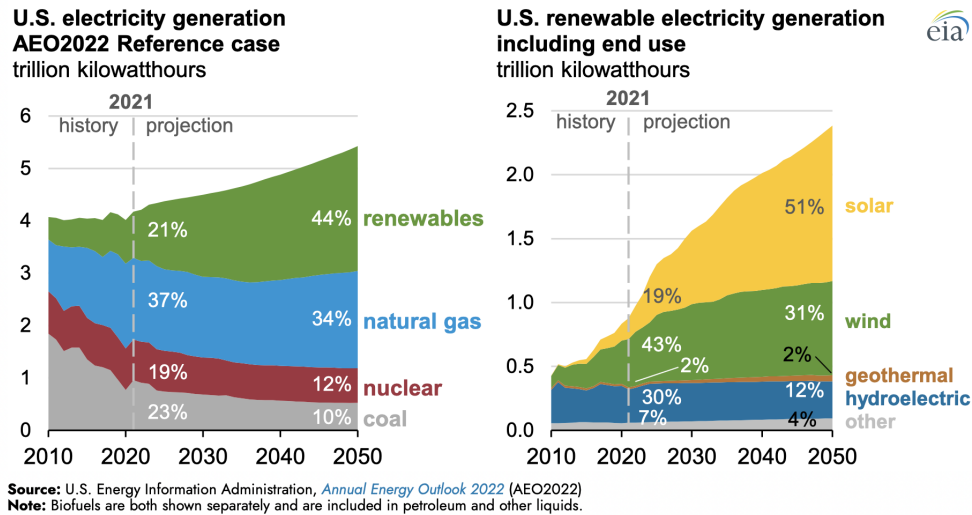


Figure 1: U.S. Electricity Generation by Fuel Type (2000–2050). Note the flat demand during the "Efficiency Plateau" (2008–2022) followed by the projected surge driven by electrification and data centers. Source: [U.S. Energy Information Administration, Annual Energy Outlook \(AEO\)](#).

2 Energy Sources: Magnitudes and Trends

- **The Current Mix (2024):** The U.S. generation profile is currently dominated by Natural Gas (43.1%) and Nuclear (18.6%). For the first time, combined Wind and Solar (17.5%) surpassed Coal (15.8%) ([Ember 2025 Report](#)).
- **The Low-Carbon Gap:** Approximately 44% of the U.S. grid is now carbon-free. However, the majority of this "clean" energy is provided by Nuclear and Hydro, which are relatively stagnant in growth compared to the 27% annual growth rate of Solar ([EIA 2025](#)).
- **Capacity Factor (CF) vs. Nameplate Capacity:** Energy delivery is governed by the Capacity Factor, defined as the ratio of actual energy produced to the maximum possible energy output.
 - **Nuclear:** 93.1% — The highest of any source, providing "firm" baseline power.
 - **Natural Gas:** 59.7% — Primarily used for load-following.
 - **Wind:** 33.2% — Geographically and weather dependent.
 - **Solar:** 23.2% — Limited by diurnal cycles and cloud cover.
- **Engineering Implication:** To replace a single 1 GW nuclear plant while maintaining the same annual TWh, a system requires ~3 GW of wind or ~4 GW of solar capacity, necessitating a significantly larger physical footprint ([MIT Climate Portal](#)).

3 Intermittency and Grid Stability

- **The Temporal Mismatch:**
 - *Diurnal:* Solar peaks at noon; demand peaks at 7:00 PM.

- *Seasonal*: Multi-day periods of low wind/sun (*Dunkelflaute*) require long-duration backup.
- **The Duck Curve**: Visualizing the **Net Load** (Total Demand – Variable Renewables).
 - **The "Belly"**: Midday overgeneration often leads to **Curtailement**—forced shutdown of renewables to prevent grid overvoltage ([CAISO 2025](#)).
 - **The "Neck"**: Evening ramping requirements in California now exceed **15,000 MW in 3 hours**, requiring fast-reacting gas peakers or massive battery fleets.

California's duck curve is getting deeper

CAISO lowest net load day each spring (March–May, 2015–2023), gigawatts

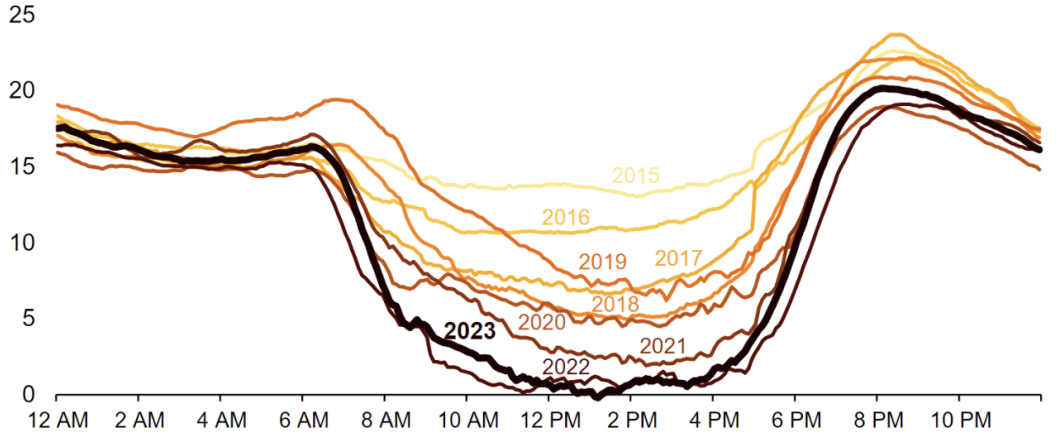


Image: Energy Information Administration

Figure 2: The CAISO "Duck Curve" showing Net Load (Total Demand minus Wind and Solar). The "belly" of the duck represents midday over-generation, while the "neck" represents the extreme ramp-up required as solar output vanishes at sunset. Source: [California Independent System Operator \(CAISO\)](#).

• The Physics of Grid Stability:

- **Rotational Inertia**: Synchronous generators (Nuclear/Fossil) provide kinetic energy that naturally stabilizes grid frequency (60 Hz).
- **Inverter-Based Resources (IBRs)**: Solar/Wind lack physical inertia. High IBR penetration leads to a high **Rate of Change of Frequency (RoCoF)**, increasing the risk of widespread blackouts during minor disturbances ([NREL 2025 Report](#)).
- **The Inertia Equation**: The relationship between power imbalance (ΔP) and frequency change (f) is governed by the Swing Equation:

$$\frac{2H}{f_0} \frac{df}{dt} = P_m - P_e$$

Where H is the inertia constant. As $H \rightarrow 0$ (high inverter penetration), the frequency gradient $\frac{df}{dt}$ (RoCoF) becomes theoretically infinite for any power mismatch, leading to instantaneous system instability unless *Synthetic Inertia* is provided via fast-acting batteries ([NERC IBR Resources](#)).

4 Storage Options and Constraints

- **Diurnal Shifting:**

- Utility-scale Li-ion is economically optimized for 4-hour "peak shaving."
- Round-trip efficiency ($\sim 85\text{--}90\%$) is high, but capacity degrades over 10–15 years.

- **The Multi-Day Problem: *Dunkelflaute***

- Defined as periods of low wind/solar lasting 5–10 days.
- These events occur 2–5 times per winter in northern latitudes ([gridX 2025 Analysis](#)).
- Storage required for a 7-day "dark lull" for a national grid is currently cost-prohibitive.

- **Scaling Physics and Material Requirements:**

- **Energy Density:** Fission energy density (7.9×10^7 MJ/kg) is approximately 8 orders of magnitude greater than electrochemical storage (0.75 MJ/kg).
- **Material Intensity:** Grid-scale batteries require massive mineral throughput. IEA projections suggest a 40x increase in Lithium demand by 2040 ([IEA 2024](#)).
- **Copper Bottleneck:** Low-density sources require extensive electrical interconnects, increasing copper usage by 400–1000% compared to centralized firm power ([IDTechEx Report](#)).

- **The Lithium Material Bottleneck:**

- *Storage Goal:* 24-hour backup for the US Grid (≈ 11 TWh).
- *Resource Intensity:* Modern LFP grid batteries require ≈ 0.1 kg of elemental lithium per kWh, which converts to **≈ 0.53 kg of Lithium Carbonate Equivalent (LCE)** per kWh.
- *Massive Scaling:* Providing just one day of backup requires **≈ 5.8 million tonnes of LCE**.
- *The Reality Check:* Total global lithium production in 2024 was ≈ 1.2 million tonnes LCE.
- **Conclusion:** Backing up the US grid for a single day would consume **100% of global lithium production for 5 years**, assuming zero production for EVs or electronics ([USGS 2025](#)).

5 Beyond Lithium: Alternative Long-Duration Storage (LDES)

While Lithium-Ion is physically and economically optimized for short-duration (4-hour) diurnal shifting, other technologies are being developed for the "Multi-Day" problem. The fundamental engineering advantage of these alternatives is the **Decoupling of Power and Energy**.

- **The Scaling Law:**

- *Lithium-Ion (Coupled):* To double the storage duration, you must duplicate the entire cell (cathode, anode, casing). The cost scales linearly with energy ($\$/\text{kWh} \approx \text{constant}$).

- *Flow/Fluid Systems (Decoupled)*: Power is set by the stack or turbine size; Energy is set by the reservoir size. To double the duration, you simply build a bigger tank. The marginal cost of energy is low.

- **Leading Candidates:**

- **Pumped Hydro**: The historical baseline ($> 90\%$ of current global storage). High efficiency ($\sim 80\%$), but geographically constrained and difficult to permit new reservoirs.
- **Flow Batteries (e.g., Vanadium Redox)**:
Uses liquid electrolytes stored in external tanks pumped through a membrane stack.
 - * *Pros*: Cycle life is effectively infinite; non-flammable.
 - * *Cons*: Lower energy density and lower Round-Trip Efficiency (RTE $\sim 70\%$) compared to Lithium.
- **Thermal Storage (Molten Salt / Carbon Blocks)**: Stores energy as heat (ΔT). Materials are cheap, but converting heat back to electricity is bound by the **Carnot Limit**, resulting in low RTE ($< 50\%$).
- **Hydrogen / Ammonia (Power-to-Gas)**: Using excess electricity to electrolyze water.
 - * *Pros*: The only viable solution for **Seasonal Shifting** (months).
 - * *Cons*: The **Round-Trip Efficiency** is poor ($\sim 30 - 40\%$). You lose the majority of your energy in the conversion steps (Electrolysis $\rightarrow$ Compression $\rightarrow$ Combustion).

6 The Grid Backbone: Firm Capacity

- **Definition of Firm Power**: Power generation that is dispatchable and guaranteed to be available at any time, for any duration, regardless of weather conditions.
- **Backbone Requirements**:
 - Reliability requires that firm capacity (Nuclear, Hydro, or Fossil with CCS) should meet at least **80–100% of the minimum seasonal peak demand** ([MIT 2018/2024 Update](#)).
 - Systems without sufficient firm backbone rely on *load shedding* (controlled blackouts) during extreme weather events.
- **Economic Thresholds and System LCOE**:
 - **The Cost Convexity**: As renewable penetration increases, the marginal cost of adding the next MWh increases exponentially due to storage and transmission requirements.
 - **Avoided Costs**: Nuclear serves as a “system-cost minimizer.” By providing firm power, it reduces the total amount of redundant “nameplate” capacity and storage the grid must maintain ([OECD-NEA System Cost Study](#)).
 - **Energy Security**: Firm capacity ensures “Black Start” capability—the ability to restart the grid from scratch after a total collapse.
- **The LCOE Paradox: Plant vs. System Costs**

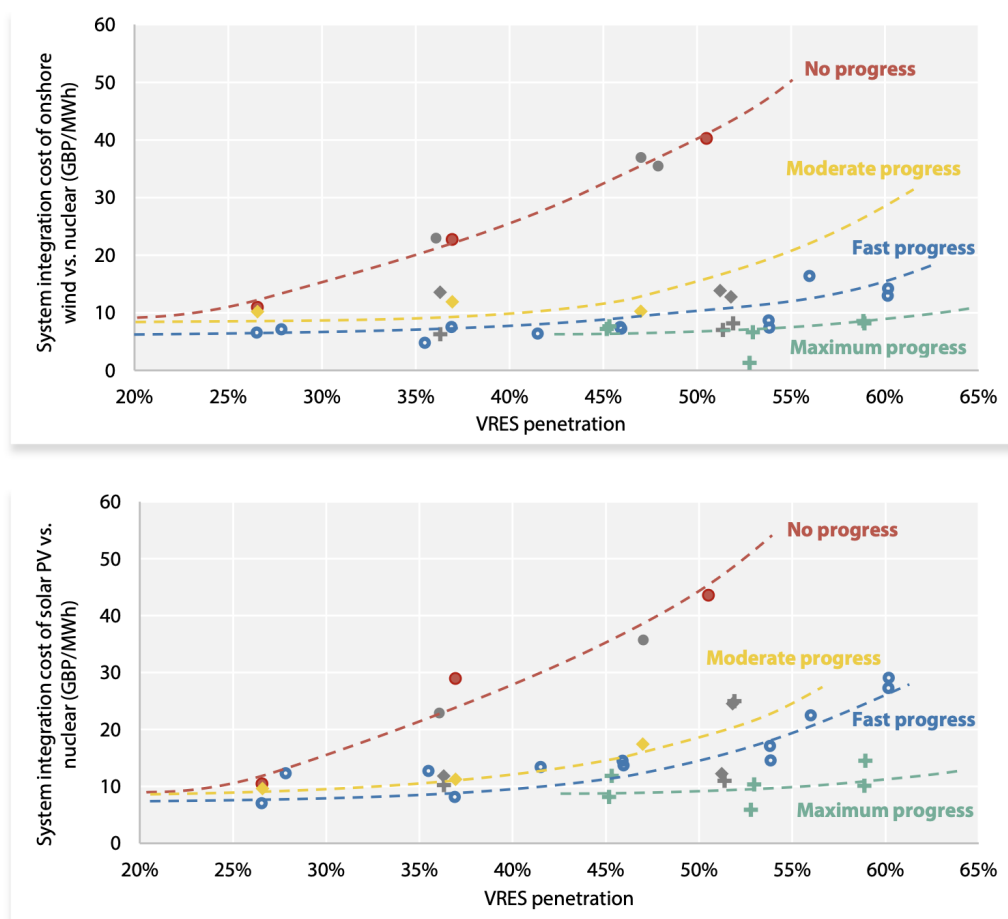
Plant-Level LCOE: The nominal cost of generation “at the busbar”:

- *Renewables (Wind/Solar)*: $\approx \$30\text{--}\$50/\text{MWh}$
- *Nuclear (New Build)*: $\approx \$70\text{--}\$90/\text{MWh}$

The System-Level Reality: As variable renewable penetration approaches 100%, "System LCOE" can exceed **$\$200/\text{MWh}$** due to the massive capital requirements for multi-day storage and grid overbuilding.

The Economic Sweet Spot: Modeling by the OECD-NEA suggests that a "firm backbone" of approximately 40% Nuclear results in the **lowest total cost to the consumer** by eliminating the most expensive 20% of required energy storage ([OECD-NEA 2019/2024](#)).

Figure 12. **System costs for wind (up) and solar PV (bottom) as a function of penetration level and progress on flexibility**



Source: Strbac and Aunedi, 2016.

Note: The different colours represent the different assumptions about the level of flexibility in the system; from "No progress" to "Maximum progress".

Figure 3: Total System LCOE vs. Variable Renewable Energy (VRE) Penetration. While the 'at-the-fence' cost of wind and solar is low, the total system cost increases exponentially as penetration exceeds 60–70% due to integration, backup, and grid costs. Source: [OECD-NEA \(2019/2024 update\)](#).

7 Fossil vs. Nuclear: The Thermal and Economic Comparison

- **The Heat Engine:** Both systems serve as heat sources for a Rankine steam cycle.
 - **Fossil:** Utilizes chemical combustion to produce steam.
 - **Nuclear:** Utilizes the kinetic energy of fission products to produce steam.
 - *Note:* While standard LWRs operate at lower temperatures ($\sim 300\text{--}315^\circ\text{C}$) due to water-coolant constraints, the "fuel" itself is not the bottleneck; the thermal limits are determined by the chosen coolant and cladding.
- **Fuel Economics and Price Sensitivity:**
 - *Fossil:* High variable costs; fuel is **70–80%** of O&M. Operation is sensitive to commodity market fluctuations.
 - *Nuclear:* Low variable costs; fuel is **$\sim 15\%$** of O&M. The cost of electricity is largely fixed by the initial capital investment ([NEI 2025](#)).
- **Waste Streams and Externalities (per 1,000 MWe per Year):**
 - *Fossil (Coal):* $\sim 300,000$ tonnes of solid ash; $\sim 6,000,000$ tonnes of gaseous CO_2 released or sequestered ([WNA 2024](#)).
 - *Nuclear:* $\sim 25\text{--}30$ tonnes of solid spent fuel.
 - **Containment Philosophy:** Nuclear relies on a "Capture and Contain" model; fossil historical relies on "Dilute and Disperse."

8 Course Roadmap: From Physics to Systems

- **Unit I: The Physics of the Nucleus (Weeks 1–4)**
 - Fundamental particles, binding energy, and mass defect.
 - Radioactive decay kinetics and ionizing radiation.
- **Unit II: The Neutron and the Chain Reaction (Weeks 5–9)**
 - Interaction probabilities (Cross-sections, σ).
 - Moderation, Diffusion Theory, and the Six-Factor Formula for Criticality (k_{eff}).
- **Unit III: Reactor Engineering & Safety (Weeks 10–14)**
 - Thermal-hydraulics, reactor designs (PWR, BWR, SMR), and the fuel cycle.

Additional References

1. **Boston University US Energy Charts** A nice visual guide to US electrical energy usage and sources over time.
<https://visualizingenergy.org/united-states-electricity-history-in-four-charts/>